Federal Aviation Administration FAARFIELD Detailed Computation Report

FAARFIELD 2.1.1 (Build 03/15/2026)

Job Name: New Job 1

Structure: New Structure 1

Analysis Type: HMA on Aggregate

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MAX TOTAL CDF	DESIGN THICKNESS	AIRCRAFT IN MIX	CRITICAL OFFSET
0.999945	10.93 in.	2	200 in.

SUBGRADE MODULUS	CONVERGED / ITERATIONS
15,000 psi	Yes / 4

Table of Contents

A. Pavement Structure Summary

Layer #	Thickness (in.)	Modulus (psi)	LCode
1	4.00	200,000	1
2	17.45	74,897	6
3	10.93	20,691	8
4	Semi-infinite	15,000	4

Expanded Sublayer Structure (after modulus adjustment)

Sublayer #	Thickness (in.)	Modulus (psi)	LCode
1	4.00	200,000	1
2	7.45	90,036	6
3	10.00	63,620	6
4	10.93	20,691	8
5	Semi-infinite	15,000	4

Evaluation Depth at Subgrade = 32.38 in.

B. Design Equations

Subgrade Strain Failure Model (FAA Standard)

$$AA = 0.000247 + 0.000245 \times \log_{10}(E\_subgrade)$$
$$BB = 0.0658 \times E\_subgrade^{0.559}$$
$$N\_fail = 10{,}000 \times (AA / \varepsilon v)^{BB}$$

For this structure (E_subgrade = 15,000 psi):

$AA = 0.001270$

$BB = 14.212$

Cumulative Damage Factor (CDF)

$$CDF\_aircraft = Repetitions \times (C/P) / N\_fail$$

$$CDF\_total = \Sigma\, CDF\_aircraft \quad \text{(summed over all aircraft)}$$

where C/P = Coverage-to-Pass ratio from Gaussian lateral wander model ($\sigma = 30.435$ in.)

Coverage-to-Pass (Gaussian Wander Model)

$$C/P(offset) = \int G(x; \sigma)\, dx \quad \text{over tire contact width}$$

$$G(x; \sigma) = [1 / (\sigma\sqrt{(2\pi)})] \times \exp(-x^2 / 2\sigma^2)$$

$\sigma = 30.435$ in. (std. dev. of lateral wander)

Projected tire width at depth d: $TW\_proj = TW + 2d$

Convergence Criterion

$$|\ln(CDF\_total)| < 0.005 \quad \text{(exit criterion)}$$

Design converges when $0.995 < CDF\_total < 1.005$

Sublayers activated when $0.5 < CDF < 2.0$ ($|\ln(CDF)| < 0.690$)

C. Understanding Coverage-to-Pass (C/P)

The Coverage-to-Pass (C/P) ratio is the probability that a given evaluation strip (10 in. wide) on the pavement surface will be loaded by a passing wheel, accounting for Gaussian lateral wander of the aircraft about the nominal wheel path centerline. The standard deviation of wander is σ = 30.435 in. (corresponding to a wander width of 70 in.). For multi-wheel gear configurations, the total C/P at each strip is the superposition (sum) of the individual GaussArea contributions from every wheel in the gear assembly, because the entire gear wanders as a rigid body. FAARFIELD evaluates 41 strips on one side of the nominal wheel path centerline (offsets 0 to 400 in.), exploiting symmetry of the Gaussian distribution.

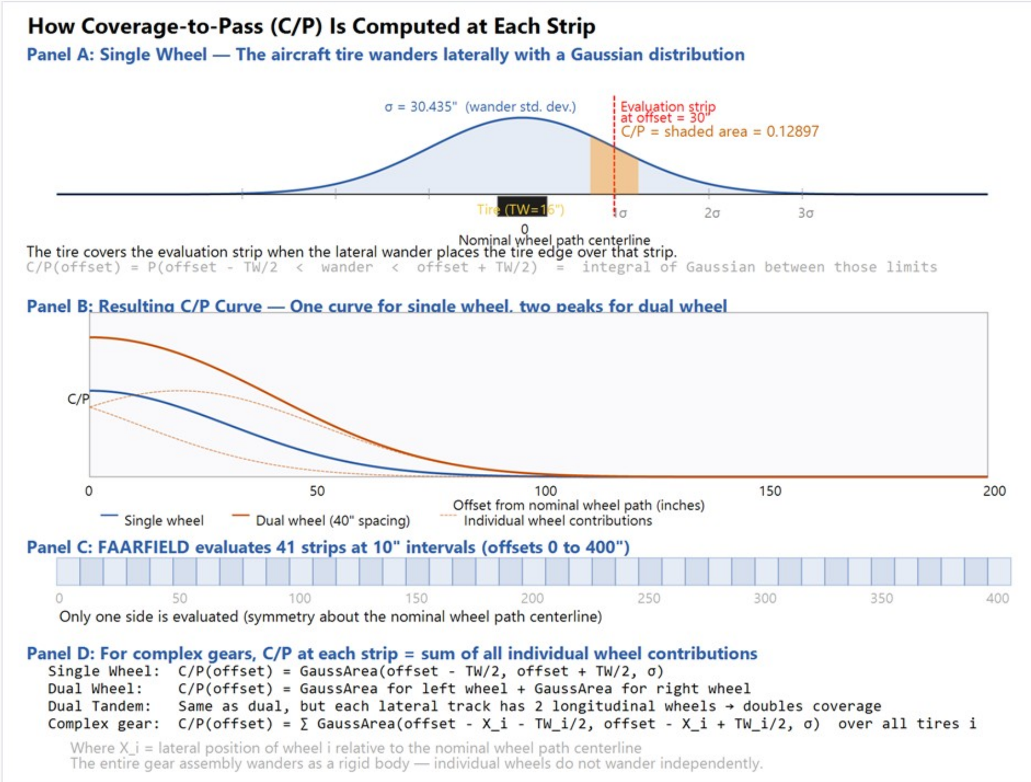


Figure: Multi-panel educational diagram showing how C/P is computed. Panel A: Gaussian wander with a single tire. Panel B: C/P curves for single vs. dual wheel configurations showing individual wheel decomposition. Panel C: The 41 evaluation strips. Panel D: Superposition formula for multi-wheel gears.

D. Fatigue Characterization

The following plot shows the subgrade fatigue model curve (allowable repetitions vs. vertical strain) with each aircraft's computed strain and N_fail plotted as scatter points. Horizontal dashed lines indicate the total repetitions (design traffic) for each aircraft. The margin between the scatter point and the dashed line represents the available fatigue life reserve.

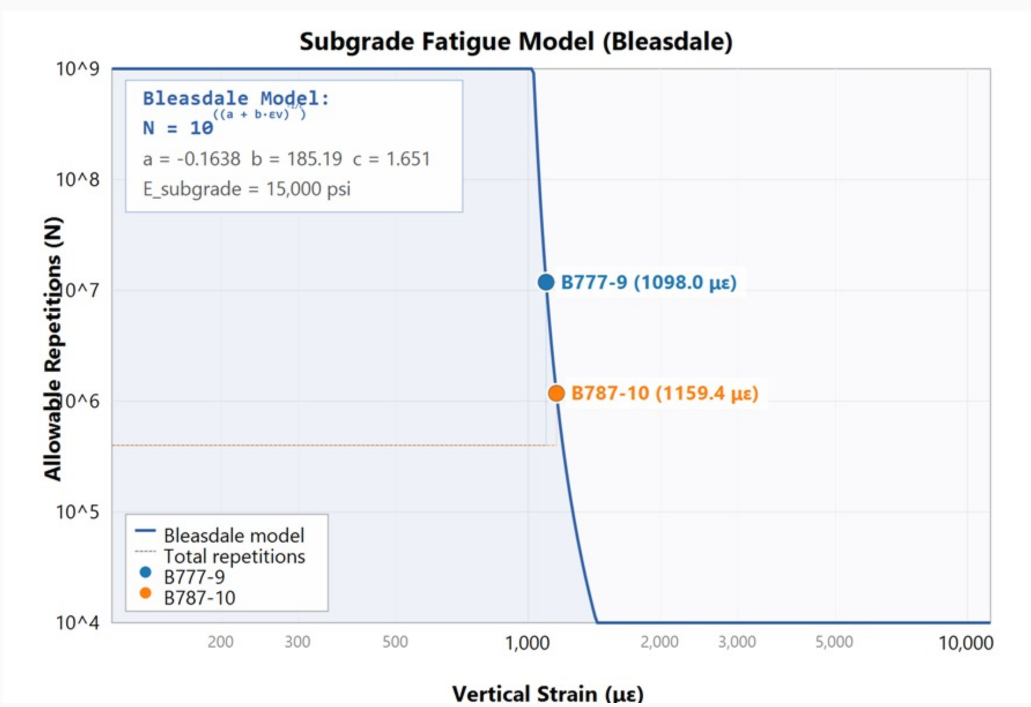


Figure: Subgrade fatigue model with aircraft operating points and design traffic levels.

Aircraft Fatigue Parameters

Aircraft	Vert. Strain (microstrain)	AA	BB	N_fail	Repetitions	N_fail / Reps	Model
B777-9	1098.03	0.002428	14.212	1.189E+07	400,000	2.97E+01	Bleasdale
B787-10	1159.38	0.002428	14.212	1.181E+06	400,000	2.95E+00	Bleasdale

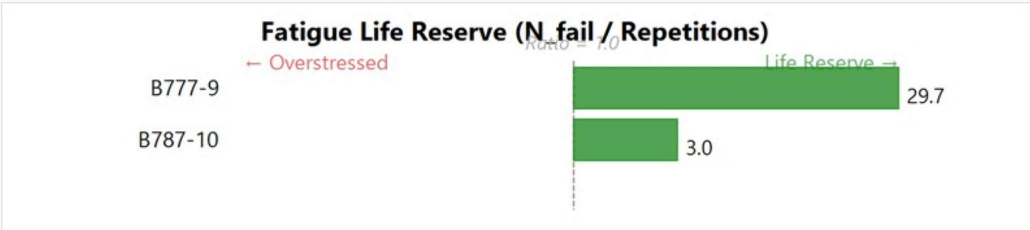


Figure: Fatigue life reserve for each aircraft. Green bars (right of center) indicate life reserve (N_fail > Repetitions). Red bars (left of center) indicate the aircraft exceeds its fatigue life at this thickness. The reference line at ratio = 1.0 represents exact balance between design traffic and allowable repetitions.

E. Per-Aircraft Detailed Breakdown

Aircraft 1: B777-9

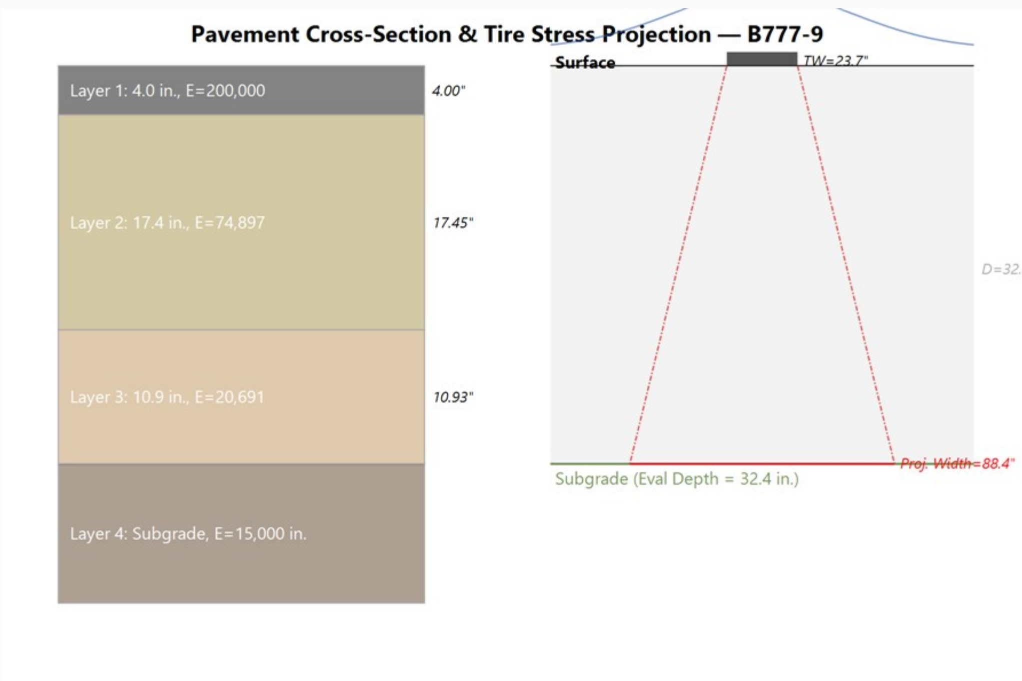


Figure: Pavement cross-section with stress projection for B777-9. Orange/brown dashed lines show the 45° stress spread from tire edges. Blue curve shows Gaussian lateral wander distribution (σ=30.435 in.).

Parameter	Value	Description
Gear Type	X	Landing gear configuration
Gear Load	777,000 lbs	Maximum gear load applied to pavement
Tire Pressure	14.8 psi	Inflation pressure
Tire Width (TW)	23.65 in.	Contact width at surface
Contact Area	0.00 in.^{2}	Static tire contact area
Annual Departures	10,000	Annual departure level (ADL)
Total Repetitions	400,000	Computed from ADL × 20-year design life
Number of Gear Loads	1	Main + belly gear load groups
Projected TW at Subgrade	88.41 in.	TW + 2×depth (45° spread)
Max Vertical Strain	1098.03 με	LEAF-computed subgrade strain
N_fail	1.189E+07	Allowable load repetitions (Bleasdale)
Max C/P Ratio	0.80318	Peak coverage-to-pass ratio
Max CDF (this aircraft)	0.496982	Peak damage contribution
CDF at Critical Offset	0.484729	Damage at the critical strip location

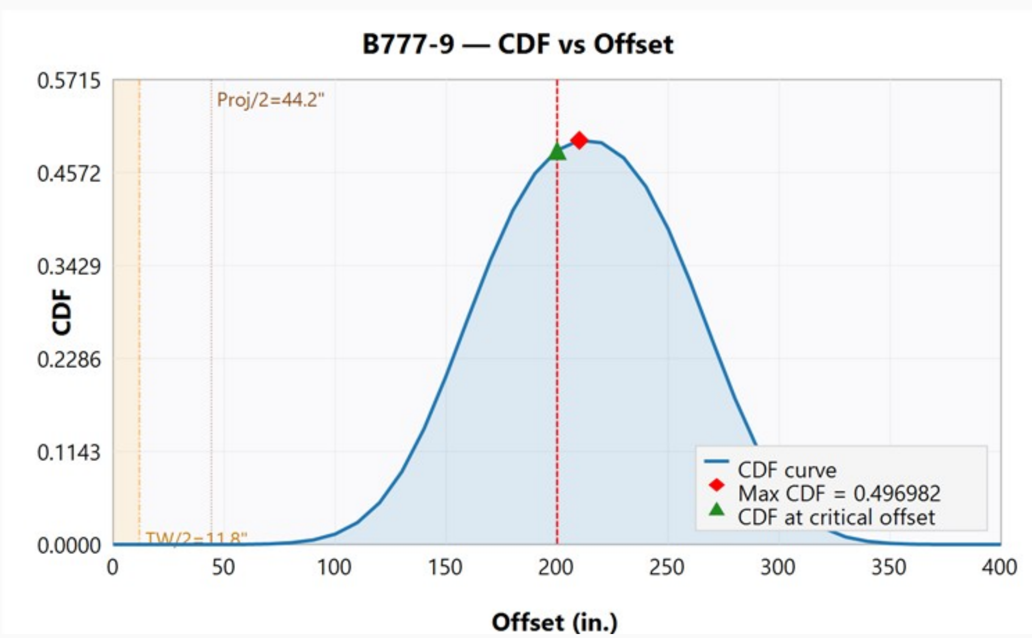


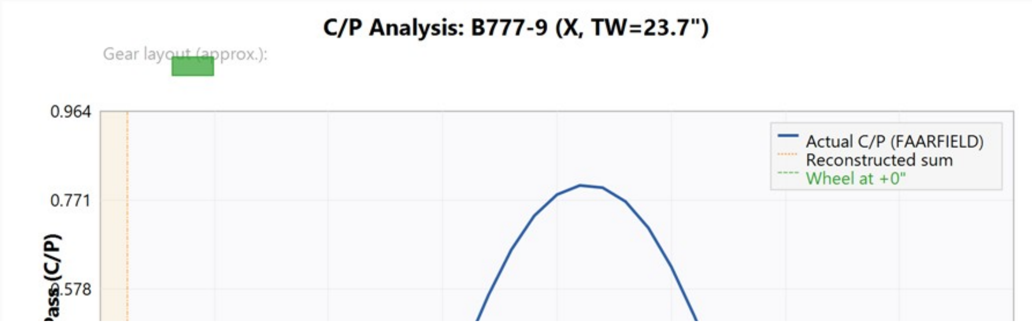
Figure: CDF distribution for B777-9. Orange band = half tire width zone. Brown dashed = half projected width at subgrade. Red dashed = critical strip. Red diamond = max CDF point. Green triangle = CDF at critical offset.

CDF by Offset — B777-9

Offset (in.)	C/P	CDF
0	0.00000	0.000000
10	0.00000	0.000002
20	0.00000	0.000002
30	0.00001	0.000006
40	0.00004	0.000023
50	0.00013	0.000081
60	0.00043	0.000263
70	0.00129	0.000799
80	0.00358	0.002213
90	0.00905	0.005601
100	0.02085	0.012900
110	0.04366	0.027017
120	0.08321	0.051487
130	0.14460	0.089474
140	0.22982	0.142207
150	0.33542	0.207544
160	0.45186	0.279596
170	0.56570	0.350039
180	0.66378	0.410724
190	0.73730	0.456215
200	**0.78338**	**0.484729**
210	0.80318	0.496982
220	0.79828	0.493950
230	0.76821	0.475346
240	0.71115	0.440033
250	0.62714	0.388051
260	0.52138	0.322614
270	0.40480	0.250474
280	0.29122	0.180197
290	0.19296	0.119395
300	0.11719	0.072516
310	0.06501	0.040228
320	0.03286	0.020332
330	0.01510	0.009346
340	0.00631	0.003907
350	0.00240	0.001488
360	0.00084	0.000520
370	0.00027	0.000167
380	0.00008	0.000050
390	0.00002	0.000014
400	0.00001	0.000004

Wheel-Level C/P Breakdown — B777-9

> The following chart decomposes the total C/P curve into individual wheel contributions for B777-9 (gear type: X). Each wheel in the gear contributes a bell-shaped GaussArea curve centered on its lateral position relative to the nominal wheel path centerline. The total C/P at any strip is the sum of all individual wheel contributions, because the entire gear assembly wanders as a rigid body.


C/P Analysis: B777-9 (X, TW=23.7")
Gear layout (approx.):
0.964
0.771
0.578
Pass (C/P)
Actual C/P (FAARFIELD)
Reconstructed sum
Wheel at +0"